Compute and servers

What is a virtual priv

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What is a VPC?

A virtual private cloud (VPC) is a publ
enterprise establish its own private c
shared public cloud infrastructure.

A VPC allows an enterprise to define and cont
from all other public cloud tenants, creating a

Imagine that a cloud provider's infrastructure
multiple families living inside. Being a public c
with a few roommates. In contrast, having a V
condominium—no one else has the key, and n
permission.

A VPC's logical isolation is implemented by us
features that give an enterprise customer grar
applications can access particular resources.
only" or "public/private" controls on social me
see your otherwise public posts.

VPC falls under the infrastructure as a service
cloud service offerings, along with platform as
(SaaS) and serverless. All top cloud service pr
Web Services (AWS), Microsoft Azure, Google
VMware and more.

Industries that require high levels of security,
healthcare, finance and government, often fav
Insights, Inc. report, the virtual private cloud (
USD 38.8 billion in 2022 to USD 129.6 billion
include the rising demand for simple installati
solutions and the growing adoption of virtual p
businesses.[1]

Check out this video with Ryan Sumner from I
architecture and its benefits.

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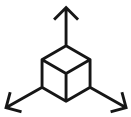
What is a Virtual Private Cloud? (4:50 min)

Features of a VPC

VPCs are a "best of both worlds" approach to
advantages of private clouds while using publi
are some key features of the VPC model.

Agility
Control the size of your virtual ne
whenever your business needs t
dynamically and in real time.



Availability
Redundant resources and highly
architectures mean that your ap
available.

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Security
Because a VPC is a logically isola
won't share space or mix with th
customers. You have complete c
are accessed and by whom.

Affordability
VPC customers can take advanta
effectiveness, such as saving on
resources.

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Benefits of a VPC

Each VPC's main features readily translate int
agility, increased innovation and faster growth

- **Flexible business growth:** Because cloud
servers, cloud storage and networking—ca
can quickly adapt to changes in business n

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- **Satisfied customers:** In today's "always-c
  expect uptime ratios of nearly 100%. The l
  reliable online experiences that build custo

- **Reduced risk across the entire data lifec**
  instance or subnet level (or both). This feat
  increases your customers' trust.

- **More resources to channel toward busine**
  demands on your internal IT team, you car
  exercising core competencies.

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VPC versus ...

VPC versus virtual privat

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A virtual private network (VPN) makes a conne
connection to a private network by creating ar
information travels. You can deploy a VPN as a
secure site-to-site communication channel be
environment or other location. By using a VPN
so that they function as if they were on a singl

VPC versus private cloud

Private cloud and virtual private cloud are som
interchangeably. A VPC is actually a *public clo*
cloud environment owned, operated and man
commonly on premises or in a dedicated spac
multi-tenant architecture, but each customer'
from those of all other tenants. The cloud prov
isolation.

VPC versus public cloud

A VPC is a single-tenant concept that allows y
cloud's architecture. A VPC offers greater secu
offerings but still lets customers take advanta
flexibility and cost-effectiveness. Sometimes,
and a public cloud account. For instance, extra
blocks of a specific size for VPCs. Not all publi
offerings.

VPC architecture

In a VPC, you can deploy cloud resources—ref
isolated virtual network. These cloud resource

- **Compute:** Virtual server instances (VSIs, a
  the user as CPUs (vCPUs) with a predetern
  and so on.

- **Storage:** VPC customers are typically alloc
  account, with the ability to purchase more.
  extra hard disk space. Storage recommenc
  workload.

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- **Networking:** You can deploy virtual versio
  virtual private cloud account to enable or r
  - **Public gateways**: Public gateways ar
    VPC environment can be made availa
  - **Load balancers**: Load balancers distr
    to optimize availability and performar
  - **Routers:** Routers direct traffic and en
    segments.
  - **Direct or dedicated links:** Direct or d
    and secure communications between
    environment or your private cloud an

Other VPC compone

- **Regions:** Providers host VPCs across regio
  where apps, services and other resources
  more zones, which are physical data cente
  storage resources, with related cooling an
  Zones are isolated from each other, which
  within a region occurs.

- **Availability zones:** An availability zone is a
  within a VPC region with independent pow

- **Subnets:** A subnet is a logical partition of a
  network segments. These fundamental me
  to individual resources (like virtual server i
  these resources through network access c
  resource groups. In a VPC environment, su
  cannot be accessed publicly through the Ir

- **Route tables:** Each subnet in a VPC must b
  of rules or routes that control network traf

- **Flow logs**: Flow Logs enable the collection
  about the IP traffic going to and from netw

- **Domain name system (DNS) services:** DN
  to create their own private DNS zones and
  improve online privacy and security by enc
  parties from monitoring online activity.

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Three-tier architectu

Most of today's software applications are desi
of the following interconnected tiers.

Web/presentation tier
The web or presentation tier tak
presents information created by,
users. This top-level tier can run
application) or a graphical user i

Application tier
The application tier, sometimes
business logic and is where mos

Database tier
The database tier comprises clo
in the application tier.

In a three-tier application, all communication
presentation and data tiers cannot communic
tier communicates with the presentation and
interface (API) calls.

To create a three-tier application architecture
subnet, giving it its own IP address range. Eac
unique ACL.

VPC security

In a virtual private cloud (VPC) model, the VP
remains isolated and secure. They accomplish
technologies, including network isolation—su

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local area networks (VLANs) and so on—that h
traffic.

Also, VPCs achieve high levels of security by c
features used to control access to resources h
security features enable customers to define v
the public cloud and control which IP address

Two types of network access controls compris

- **Access control lists (ACLs):** An ACL is a lis
particular subnet within your VPC. As previ
subdivision of your VPC; the ACL defines th
access to it.

- **Security group:** With a security group, you
be situated in more than one subnet) and a
example, if you have three applications in t
be public Internet-facing, you can place th
groups act like virtual firewalls, controlling
matter which subnet they are in.

VPC pricing

The various cloud providers can offer different
common for individual VPC resources—such a
priced separately. Data transfer charges are al
cloud providers do not charge for data transfe

Determining the best VPC and pricing model t
considering the requirements of the applicatic
compute-intensive? Will they require large am
balanced regarding their CPU, storage and me
questions will help predict your usage needs,
when comparing options.

VPC use cases

- **Host web applications:** Securely host web
how network traffic can reach your VPC res

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- **Cloud migration:** VPC provides a cost-effic
an isolated private cloud within a public cl
minimal downtime and robust cloud securi

- **Hybrid cloud strategy:** A VPC supports toc
connect VPCs to a public cloud or on-prem
prem, private and public cloud resources t
infrastructure.

- **Multicloud deployment**: VPCs also suppo
connections between VPCs across cloud p
source, cloud-native technologies like Kub
capabilities for managing workloads acros
or single pane of glass.

- **DevOps practices:** VPC environments sup
delivery of higher-quality applications and
native tools and technologies to perform r
the entire software development lifecycle.

- **High-performance computing (HPC):** VPC environments offer fast-provisioning compute capacity with the highest networking speeds and the most secure, software-defined networking resources to support high-performance computing (HPC) needs of highly regulated industries like finance, healthcare and more.

- **Regulatory compliance and data governance:** Adhering to strict regulatory and data governance requirements is critical, especially for global industries like oil and gas. Today's managed service providers for cloud VPCs offer built-in security and regulatory compliance tools and hardware and software solutions for confidential computing. Standard features include encryption options and data residency controls.

- **Industry-specific clouds:** VPCs are an ideal component of industry-specific cloud platforms, a growing trend among industries like finance and healthcare that are looking for sector-specific capabilities that are secure and can deliver business outcomes faster

- **Business continuity disaster recovery (BCDR):** Like other cloud-based services, business continuity disaster recovery (BCDR) with VPCs involves mechanisms to protect data and restore service functions. These incl ures
to restore a system, application or data center ial
infrastructure in a VPC across different region e
event of a disaster (for example, equipment fa).

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- **Edge computing and the Internet of Things (IoT):** As more industries like manufacturing and retail use IoT and edge devices connect to the cloud, there is a r for secure and scalable cloud environments like VPCs.

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Take the next step

You've explored the foundations - now take a closer look at how organizations are applying GenAI in real environments, supported by the right mix of cloud infrastructure and networking architecture to scale responsibly.

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